Tutorial: testing mediation effect for sparse microbiome data using CRAmed

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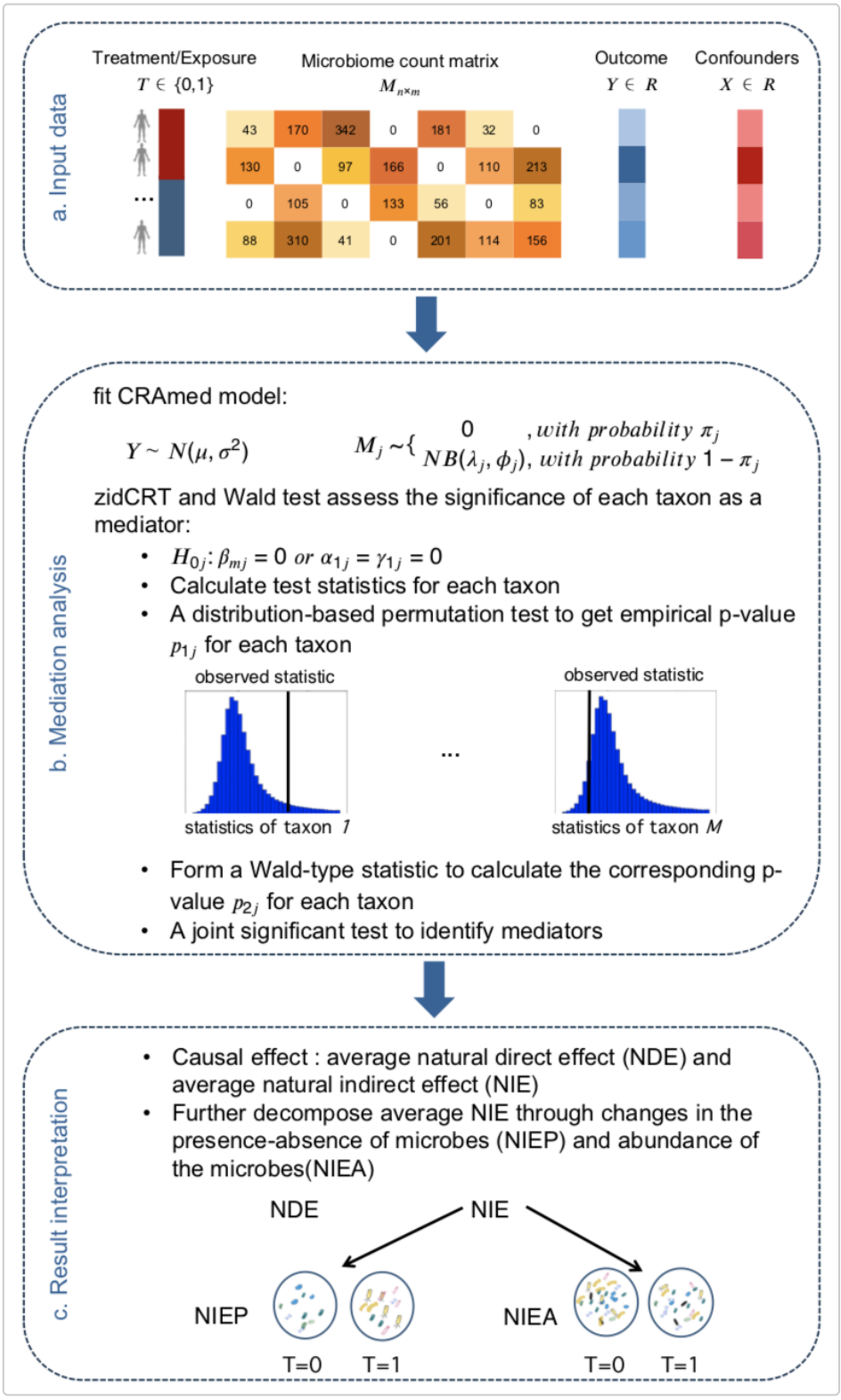
This is a practical tutorial on the use of CRAmed package, which introduces a mediation test for sparse and high-dimensional microbiome data.

A brief summary of the CRAmed

CRAmed uses a zero-inflated negative binomial (ZINB) distribution to model the microbiome count data obtained from 16S rRNA and metagenomic shotgun sequencing. The ZINB model naturally decomposes the total natural indirect effect (NIE) into two components. The first component, rep- resenting variations in the abundance of taxa, is captured by the negative binomial part of the model. The second component, representing the influence of the presence-absence of taxa, is captured by the zero-inflated part of the model. To account for the high dimensionality of microbiome data, CRAmed introduces a zero-inflated conditional randomization test (zidCRT). This approach enables us to identify the relevant taxa that act as causal mediators while controlling the false discovery rate. Finally, a joint significant test is emploied to identify the significant mediators.



A schematic and outline of the CRAmed methodology for microbiome data.

The overall workflow of CRAmed is shown in the figure above.

Application of CRAmed: Infant data

The composition of newborns' microbiome undergoes significant influence based on the mode of delivery (Mitchell et al. (2020)). Several studies have demonstrated that cesarean section (C-section) disrupts the succession of the newborn microbiome from the maternal birth canal and increases the risk of adverse health outcomes in offspring compared with vaginally delivered infants (Andersen et al. (2020),Korpela et al. (2020),Zhou et al. (2023)). Therefore, it is crucial to understand how the delivery mode influences the infant gut microbiome, which further mediates the phenotype of infants.

To identify the gut microbiome mediator between delivery mode and the infants' weight, we apply our CRAmed and other methods to a publicly available dataset, coined DIABIUMMUNE (Yassour et al. (2016)).

```
library(CRAmed);library(phyloseq);library(pscl);#packageVersion("pscl") '1.5.5'
#> Classes and Methods for R originally developed in the
#> Political Science Computational Laboratory
#> Department of Political Science
#> Stanford University (2002-2015),
#> by and under the direction of Simon Jackman.
#> hurdle and zeroinfl functions by Achim Zeileis.
# Load data
data(Infant)
# remove missing values
Infant_sub <- subset_samples(Infant, !is.na(weight_growth_pace_during_three_years))
# filter out taxa with a prevalence of less than 10\%
prevdf <- apply(X = otu_table(Infant_sub),
                MARGIN = ifelse(taxa_are_rows(Infant_sub), yes = 1, no = 2),
                FUN = function(x){sum(x > 0)})
prevdf <- data.frame(Prevalence = prevdf,
                     TotalAbundance = taxa_sums(Infant_sub),
                     tax_table(Infant_sub))
prevalenceThreshold = floor(0.1* nsamples(Infant_sub))
keepTaxa <- rownames(prevdf)[(prevdf$Prevalence >= prevalenceThreshold)]
filter.infant <-  prune_taxa(keepTaxa, Infant_sub)
# metadata of Infant data
meta.infant <- sample_data(filter.infant)
# microbiome data
M_mat <- t((otu_table(filter.infant)@.Data))
# outcome
Y <- as.numeric(meta.infant$weight_growth_pace_during_year_one)
#treatment
trt <- ifelse(meta.infant$csection == TRUE, 1, 0)
```

To run `CRAmed` function, the minimum requirement is to provide `treatment`, `mediators`, and `outcome`. Note that if `CI=TRUE` and `n.perm=100` the function will cost several minutes to output the p-values calculated through permutation procedure. In order to save time, in the chunk below, we set `FDR = 0.05`, `n.times=10` (the number of times for sampling mediators from the ZINB model), `n.perm=10`(the number of times for sampling dataset for calculating confidence interval) and `CI=TRUE` in identifying mediators.

```
results.infant <- try(CRAmed(M_mat=as.matrix(M_mat), Y=Y, Exposure=as.matrix(trt), FDR =
        0.05, n.times=10, n.perm=10, CI=TRUE))
#> 载入需要的程辑包：Matrix
#> Loaded glmnet 4.1-6
#> [1] "############# d0CRT pvalue ###############"
#> [1] "6: 0.272727272727273"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "17: 0.181818181818182"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "24: 0.0909090909090909"
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#> [1] "46: 0.0909090909090909"
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#> [1] "############# d0CRT pvalue ###############"
#> [1] "95: 0.363636363636364"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "102: 0.0909090909090909"
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#> [1] "131: 0.272727272727273"
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#> [1] "768: 0.272727272727273"
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#> [1] "873: 0.363636363636364"
#> [1] "ZINB"
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#> [1] 2
#> [1] 3
#> [1] 4
#> [1] 5
#> [1] 6
#> [1] 7
#> [1] 8
#> [1] 9
#> [1] 10
results.infant
#> $Mediators
#> integer(0)
#>
#> $NDE
#> [1] 0
#>
#> $NIE
#> [1] NA
#>
#> $NIEA
```

```
#> [1] NA
#>
#> $NIEP
#> [1] NA
#>
#> $NDE.pval
#> [1] 0.6363636
#>
#> $NIE.pval
#> [1] NA
#>
#> $NIEA.pval
#> [1] NA
#>
#> $NIEP.pval
#> [1] NA
#>
#> $NDE.CI
#>  2.5% 97.5%
#>     0     0
#>
#> $NIE.CI
#> [1] NA
#>
#> $NIEA.CI
#> [1] NA
#>
#> $NIEP.CI
#> [1] NA
```

The output consists of several components:

- `Mediators`: identified mediators
- `NDE, NIE, NIEA, and NIEP`: corresponding natural direct effect, natural indirect effect, natural indirect effect mediated by the changes in the abundance, natural indirect effect mediated by the the presence status.
- `NDE.pval, NIE.pval, NIEA.pval, and NIEP.pval`: the corresponding p-values of NDE, NIE, NIEA, and NIEP.
- `NDE.CI, NIE.CI, NIEA.CI, and NIEP.CI`: the corresponding confidence interval of NDE, NIE, NIEA, and NIEP.

Application of CRAmed: GGMP data

Numerous studies have underscored the profound influence antibiotics can exert on the composition of bacteria within the gut microbiome (Cho et al. (2012),Fishbein, Mahmud, and Dantas (2023)). We adopted a dataset from the Guangdong Gut Microbiome Project (GGMP), a large community-based cross-sectional cohort conducted between 2015 and 2016 including 7009 participants with high-quality gut microbiome data (He et al. (2018)). After preprocessing, we apply CRAmed to test whether the gut microbiota is a mediator that mediates the association between antibiotics and CMD-related risk factors.

```
library(CRAmed)
# Load data
data(GGMP)
# filter out taxa with a prevalence of less than 10\%
prevdf <- apply(X = otu_table(GGMP),
```

```
                 MARGIN = ifelse(taxa_are_rows(GGMP), yes = 1, no = 2),
                 FUN = function(x){sum(x > 0)})
prevdf <- data.frame(Prevalence = prevdf,
                     TotalAbundance = taxa_sums(GGMP),
                     tax_table(GGMP))
prevalenceThreshold <- floor(0.1* nsamples(GGMP))
keepTaxa <- rownames(prevdf)[(prevdf$Prevalence >= prevalenceThreshold)]
filter.ggmp <- prune_taxa(keepTaxa, GGMP)
# metadata of GGMP data
meta.ggmp <- sample_data(filter.ggmp)
# microbiome data
M_mat <- t((otu_table(filter.ggmp)@.Data))
# outcome
Y <- as.numeric(unlist(meta.ggmp[,"BMI"]))
# treatment
trt <- ifelse(meta.ggmp$Antibiotics == "n", 1, 0)
```

To run `CRAmed` function, we also provide `treatment`, `mediators`, and `outcome`.

```
results.ggmp <-try(CRAmed(M_mat=as.matrix(M_mat[,1:20]), Y=Y, Exposure=as.matrix(trt),
                     FDR = 0.05, n.times=10, n.perm=10, CI=TRUE))
#> [1] "############# d0CRT pvalue ###############"
#> [1] "3: 0.0909090909090909"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "5: 0.0909090909090909"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "8: 0.636363636363636"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "9: 0.0909090909090909"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "10: 0.272727272727273"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "11: 0.0909090909090909"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "12: 0.636363636363636"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "15: 0.363636363636364"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "17: 0.272727272727273"
#> [1] "############# d0CRT pvalue ###############"
#> [1] "20: 0.0909090909090909"
#> [1] "ZINB"
#> [1] "ZINB"
#> [1] "ZINB"
#> [1] "ZINB"
#> [1] "ZINB"
#> [1] "ZINB"
#> [1] "ZINB"
#> [1] "ZINB"
#> [1] "ZINB"
#> [1] 1
#> [1] 2
#> [1] 3
#> [1] 4
```

```
#> [1] 5
#> [1] 6
#> [1] 7
#> [1] 8
#> [1] 9
#> [1] 10
results.ggmp
#> $Mediators
#> integer(0)
#>
#> $NDE
#> [1] 0
#>
#> $NIE
#> [1] NA
#>
#> $NIEA
#> [1] NA
#>
#> $NIEP
#> [1] NA
#>
#> $NDE.pval
#> [1] 1
#>
#> $NIE.pval
#> [1] NA
#>
#> $NIEA.pval
#> [1] NA
#>
#> $NIEP.pval
#> [1] NA
#>
#> $NDE.CI
#>      2.5%     97.5%
#> 0.0000000 0.1804107
#>
#> $NIE.CI
#> [1] NA
#>
#> $NIEA.CI
#> [1] NA
#>
#> $NIEP.CI
#> [1] NA
```

Andersen, Vibeke, Sören Möller, Peter Bjødstrup Jensen, Frederik Trier Møller, and Anders Green. 2020. "Caesarean Delivery and Risk of Chronic Inflammatory Diseases (Inflammatory Bowel Disease, Rheumatoid Arthritis, Coeliac Disease, and Diabetes Mellitus): A Population Based Registry Study of 2,699,479 Births in Denmark During 1973–2016." *Clinical Epidemiology* 12: 287–93.

Cho, Ilseung, Shingo Yamanishi, Laura Cox, Barbara A Methé, Jiri Zavadil, Kelvin Li, and others. 2012. "Antibiotics in Early Life Alter the Murine Colonic Microbiome and Adiposity." *Nature* 488 (7413): 621–26.

Fishbein, Skye RS, Bejan Mahmud, and Gautam Dantas. 2023. “Antibiotic Perturbations to the Gut Microbiome.” *Nature Reviews Microbiology* 21 (12): 772–88.

He, Yan, Wei Wu, Hui-Min Zheng, Pan Li, Daniel McDonald, Hua-Fang Sheng, and others. 2018. “Regional Variation Limits Applications of Healthy Gut Microbiome Reference Ranges and Disease Models.” *Nature Medicine* 24 (10): 1532–35.

Korpela, Katri, Otto Helve, Kaija-Leena Kolho, Terhi Saisto, Kirsi Skogberg, Evgenia Dikareva, and others. 2020. “Maternal Fecal Microbiota Transplantation in Cesarean-Born Infants Rapidly Restores Normal Gut Microbial Development: A Proof-of-Concept Study.” *Cell* 183 (2): 324–34.

Mitchell, Caroline M, Chiara Mazzoni, Larson Hogstrom, Allison Bryant, Agnes Bergerat, Avital Cher, and others. 2020. “Delivery Mode Affects Stability of Early Infant Gut Microbiota.” *Cell Reports Medicine* 1 (9): 100156.

Yassour, Moran, Tommi Vatanen, Heli Siljander, Anu-Maaria Hämäläinen, Taina Härkönen, Samppa J Ryhänen, and others. 2016. “Natural History of the Infant Gut Microbiome and Impact of Antibiotic Treatment on Bacterial Strain Diversity and Stability.” *Science Translational Medicine* 8 (343): 343ra81.

Zhou, Lepeng, Wen Qiu, Jie Wang, Aihua Zhao, Chuhui Zhou, Tao Sun, and others. 2023. “Effects of Vaginal Microbiota Transfer on the Neurodevelopment and Microbiome of Cesarean-Born Infants: A Blinded Randomized Controlled Trial.” *Cell Host & Microbe* 31 (7): 1232–1247.e5.